Task 4 – Best, average and worst cases

Algorithm	Best Case	Average Case	Worst Case	Input Condition
Linear Search	\(\ Theta(1)\ )	\(\Theta(n)\)	\(\Theta(n)\)	Best: first item matches. Average: target is equally likely to be at any position. Worst: target is last or absent.
Recursive Binary Search	\(\ Theta(1)\ )	\(\ Theta(\log n)\ )	\(\ Theta(\log n)\ )	Best: middle item matches immediately. Otherwise, the search interval is halved at each step until the target is found or the interval is empty.
Hash Table	Expected \(\ Theta(1)\ )	Expected \(\ Theta(1)\ )	\(\Theta(n)\)	Best/average: few collisions. Worst: many keys collide in the same bucket.